Seminar Report

A supplier selection decision model using multi-criteria decision analysis in a small manufacturing company

Review of the paper by Rodrigues, Širová and Mugurusi (2022), IFAC PapersOnLine 55(10), pp. 2773-2778, with an improved model

Submitted by:

Sujal Negi

Roll No: 123ME0023

Course Instructors:

Dr. S. Anand Kumar and ***Dr. Anuj Kumar***

Seminar (Course Code: ME496)



Department of Mechanical Engineering

Indian Institute of Information Technology, Design and Manufacturing, Kurnool

Academic Year: 2026-2027

CERTIFICATE

This is to certify that the seminar report entitled **"A supplier selection decision model using multi-criteria decision analysis in a small manufacturing company"** submitted by **Sujal Negi (Roll No: 123ME0023)** for the course **Seminar (Course Code: ME496)** at the Indian Institute of Information Technology, Design and Manufacturing, Kurnool, is a bonafide record of the work carried out by him during the Academic Year 2026-2027.

Dr. S. Anand Kumar
Course Instructor
Seminar (Course Code: ME496)
IIITDM Kurnool

Dr. Anuj Kumar
Course Instructor
Seminar (Course Code: ME496)
IIITDM Kurnool

ACKNOWLEDGEMENT

I would like to thank my course instructors, Dr. S. Anand Kumar and Dr. Anuj Kumar, for giving me the opportunity to study a research paper in depth, and for their guidance on how to read, question and improve published work.

I also thank the Department of Mechanical Engineering and the faculty members of IIITDM Kurnool for the academic environment and resources that made this seminar possible.

Finally, I thank my friends and family for their constant support and encouragement.

Sujal Negi
(123ME0023)

ABSTRACT

A small Czech company that makes car parts must buy copper rods, its most critical spare part, from one of three suppliers. The paper reviewed here asks three managers to compare six attributes in pairs on Saaty's 1 to 9 scale and solves the problem twice. The Analytic Hierarchy Process (AHP) picks Supplier 2 and the Analytic Network Process (ANP) picks Supplier 1. The authors recommend the ANP answer.

In this work every result of the paper was rebuilt from its own published tables and matched to four decimal places. The rebuilding shows that the ANP result is the same as an AHP calculation in which the quality group is made five times as important as the logistics group. The winner changes at a logistics weight of 0.442, four of the eight judgment tables fail Saaty's 5% consistency limit, and a price that is only 1.52 times higher was scored as 7.3 times worse.

An improved model is proposed that keeps the paper's input values. It re-checks one answer in each inconsistent table, scores price, delivery time and defect rate from real numbers, shows the flip point, tests the answer with 5,000 random trials and adds the expected cost of each buying plan over two years. All steps are built into a simple web tool that shows every input and calculation. Supplier 2 stays just ahead (0.374 against 0.367), but wins only 52% of the trials, so the ranking alone cannot settle the choice. Over two years Supplier 2 is the cheapest plan (€56,928 against €87,500 for Supplier 1), and Supplier 1 pays off only if one bad rod costs the company more than about €5,250.

Contents

1 Introduction **7**
- 1.1 Supplier Selection in Manufacturing . . . 7
- 1.2 The Case Company . . . 7
- 1.3 AHP and ANP in Brief . . . 7
- 1.4 Aim of This Seminar . . . 7

2 Literature Review **8**
- 2.1 Analytic Hierarchy Process (AHP) . . . 8
- 2.2 Analytic Network Process (ANP) . . . 8
- 2.3 MCDM for Spare Parts and Suppliers . . . 8
- 2.4 Supplier Selection and Order Allocation . . . 9
- 2.5 Summary of the Reviewed Literature . . . 9

3 What the Paper Did **10**
- 3.1 Decision Makers . . . 10
- 3.2 Criteria and Attributes . . . 10
- 3.3 Pairwise Judgments . . . 10
- 3.4 AHP and ANP Models . . . 11
- 3.5 Results Reported in the Paper . . . 11

4 Gaps Identified in the Paper **13**
- 4.1 How the Paper Was Checked . . . 13
- 4.2 Gap 1: Two Methods, Two Winners . . . 13
- 4.3 Gap 2: No Sensitivity Check . . . 13
- 4.4 Gap 3: Answers Not Consistent . . . 15
- 4.5 Gap 4: Measured Values Given as Opinions . . . 15
- 4.6 Gaps 5 and 6: One Fixed Ranking and No Cost View . . . 15

5 Proposed Improvement **16**
- 5.1 Objectives . . . 16
- 5.2 Re-checking Doubtful Answers . . . 16
- 5.3 Scoring from Real Numbers . . . 16
- 5.4 Measuring How Sure the Winner Is . . . 17
- 5.5 Two-year Cost of Each Buying Plan . . . 18

6 Web Tool and Results **19**
- 6.1 The Web Tool . . . 19
- 6.2 Effect of Each Improvement on the Scores . . . 19
- 6.3 How Sure the Winner Is . . . 19
- 6.4 Two-year Cost . . . 22
- 6.5 Paper's Model Compared with My Model . . . 23
- 6.6 Discussion . . . 23

7 Conclusion and Future Work **24**
- 7.1 Conclusion . . . 24
- 7.2 Recommendation to the Company . . . 24
- 7.3 Limitations of This Work . . . 25
- 7.4 Future Work . . . 25

References **26**

List of Figures

3.1 Decision tree of the paper: goal, two groups, six attributes and three suppliers 11

4.1 Final scores as the weight of the logistics group changes (paper's data) . . 14

5.1 Consistency ratio of the eight tables before and after the re-check 17
5.2 Scores on price, delivery time and part quality from opinions and from real numbers 18

6.1 Web tool: summary of the paper's model and the improved model 19
6.2 Web tool, Step 3: final score with every product of weight and score shown 20
6.3 Final scores against the weight of the logistics group in the improved model 21
6.4 Share of 5,000 trials won by each supplier 21
6.5 Where the money goes in two years for each plan 22

List of Tables

2.1 Summary of the reviewed literature 9

3.1 Group answers from the paper, recomputed weights and consistency 11
3.2 Final scores reported by the paper 12

4.1 One-attribute sensitivity at the paper's AHP setting ($w = 0.5$) 14

5.1 The one answer to re-check in each inconsistent table 16
5.2 Measured values used in the improved model 17
5.3 Inputs of the two-year cost model 18

6.1 Final scores at $w = 0.5$ as each improvement is added 20
6.2 Share of 5,000 trials in which each supplier finishes 1st, 2nd and 3rd 20
6.3 Expected two-year cost of each buying plan 22
6.4 Paper's model compared with the improved model 23

Chapter 1

Introduction

1.1 Supplier Selection in Manufacturing

In manufacturing, buying a part is not only about the lowest price. A late spare part can stop a production line, and a poor part can damage a machine or reach the customer. The decision has to balance cost, delivery time, quality, service and sustainability at the same time, which makes it a *multi-criteria decision making* (MCDM) problem.

1.2 The Case Company

The paper studies a Czech company with about 400 employees that makes components for the automotive industry. Its maintenance department buys and stores spare parts. The company faced high spare-part costs and late deliveries, and an ABC analysis showed that its most critical spare part is the copper rod: if no rod is in stock when one is needed, production stops.

Three suppliers can deliver the rod: Supplier 1 from Germany, and Suppliers 2 and 3 from the Central and North Bohemian Regions of the Czech Republic. Supplier 1 gives the best quality but has the highest price and is the farthest away. Prices range from €230 to €350 per rod, yearly spending is about €25,300, delivery takes 3 to 7 days, and the share of bad parts rose at all three suppliers from 5% to 13.33% in 2017 to 6.67% to 15% in 2018.

1.3 AHP and ANP in Brief

AHP and ANP, both developed by Saaty, break a hard decision into small questions of the form “how much more important is A than B?”. Managers answer from experience, and the method turns the answers into a score for each option. AHP arranges the problem as a tree; ANP lets its parts affect each other, like a network.

1.4 Aim of This Seminar

The aim is to understand what the paper did, check its results by rebuilding the model from the published tables, find its gaps, and propose and demonstrate an improved model that gives the company a fairer and more useful answer using the same input values.

Chapter 2

Literature Review

2.1 Analytic Hierarchy Process (AHP)

Saaty (1980) introduced AHP as a way to set out a decision as a tree: the goal at the top, criteria in the middle and options at the bottom. At each level the decision maker compares items in pairs on a scale from 1 (equally important) to 9 (extremely more important). The answers form a pairwise comparison matrix A in which $a_{ji} = 1/a_{ij}$. The weights w of the items are the principal eigenvector of A:

$$A\,w = \lambda_{\max}\,w. \tag{2.1}$$

Because people are not perfectly consistent, AHP also measures consistency:

$$CI = \frac{\lambda_{\max} - n}{n - 1}, \qquad CR = \frac{CI}{RI}, \tag{2.2}$$

where n is the size of the matrix and RI is a random index. Saaty recommends $CR \leq 5\%$ for a 3×3 matrix.

2.2 Analytic Network Process (ANP)

Saaty (2005) extended AHP to ANP. ANP drops the strict tree and lets criteria influence each other and receive feedback from the options. Items are grouped into clusters, and the influences are stored in a supermatrix that is raised to powers until it settles.

2.3 MCDM for Spare Parts and Suppliers

Gajpal, Ganesh and Rajendran (1994) and Molenaers et al. (2012) used AHP to rank spare parts by how critical they are. Kabir and Hasin (2011) showed that fuzzy AHP can capture the vagueness of human answers, and Rezaei, Fahim and Tavasszy (2014) used fuzzy AHP to select airline suppliers. Koksalmis and Kabak (2019) reviewed objective ways to give different weights to decision makers in a group.

2.4 Supplier Selection and Order Allocation

Alegoz and Yapicioglu (2019) argue that a good supplier selection method should not only pick the supplier but also decide how much to order from each one, taking cost and fast-service options into account. This supports the idea, used later in this report, that splitting an order can be a sensible choice when two suppliers are close.

2.5 Summary of the Reviewed Literature

Table 2.1: Summary of the reviewed literature

Study	**Main idea**	**What it leaves open here**
Saaty (1980)	AHP: tree, 1 to 9 scale, eigenvector weights, consistency ratio	Gives one answer with no measure of how sure it is
Saaty (2005)	ANP: network, clusters, feedback, supermatrix	Needs many extra answers; easy to use without real links between criteria
Kabir and Hasin (2011)	Fuzzy AHP compared with AHP	Not applied to this supplier case
Alegoz and Yapicioglu (2019)	Supplier selection with order allocation	Does not use managers' pairwise answers
Rodrigues et al. (2022)	AHP and ANP for a copper rod supplier	No sensitivity check, no measured data, no cost over time

Source: compiled by the author from the cited works

Chapter 3

What the Paper Did

3.1 Decision Makers

Three managers took part, because the purchase affects all three departments:

- **Maintenance manager** (4 years in the role, senior electrical engineer). Leads the purchase, negotiates price and delivery terms and controls spare-part stock.
- **Production manager** (5 years, senior mechanical engineer). Believes that a line stop costs more than any delivery charge.
- **Quality manager** (10 years, senior mechanical engineer). Follows a zero-defect policy and rates part quality far above price and delivery time.

Their individual answers were combined by the geometric mean into one group answer for every comparison.

3.2 Criteria and Attributes

Six attributes were chosen from earlier studies and placed in two groups (Figure 3.1):

- **C1 Logistic-Economic:** delivery cycle time, supplier availability (response to emergency orders) and unit price.
- **C2 Quality-Sustainability:** part quality (share of parts accepted at inspection), aftersales customer service and sustainability (share of scrapped parts the supplier remanufactures).

3.3 Pairwise Judgments

The paper prints the group answers as eight 3×3 tables (its Tables 2 to 9). Table 3.1 lists the upper half of each table (the lower half is the reciprocal), the weights recomputed in this work and the consistency ratio.

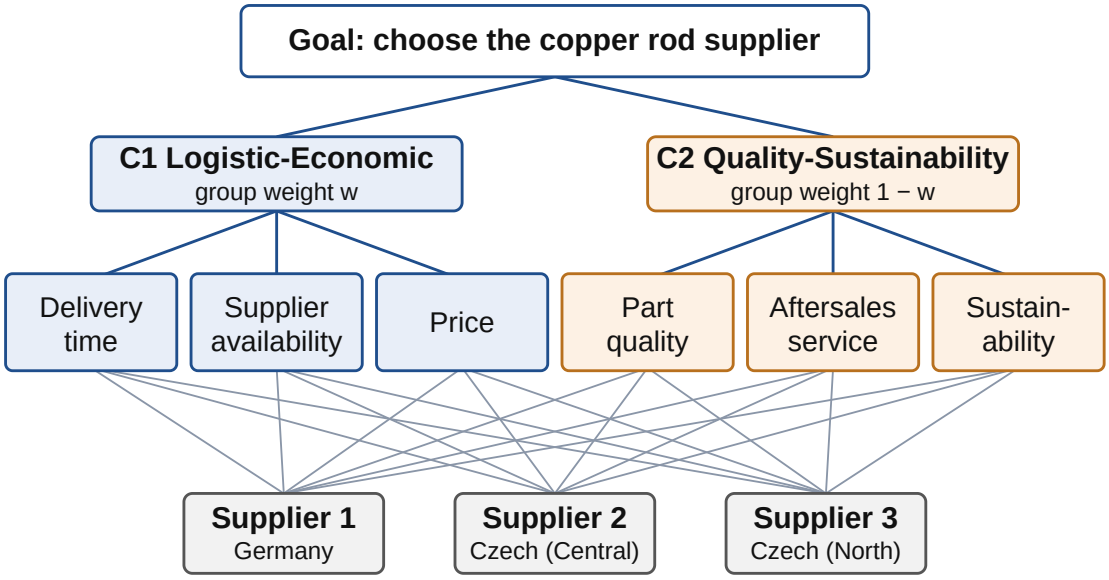


Figure 3.1: Decision tree of the paper: goal, two groups, six attributes and three suppliers

Source: redrawn from Rodrigues, Šírová and Mugurusi (2022), Figure 1

Table 3.1: Group answers from the paper, recomputed weights and consistency

Table	Items 1, 2, 3	a_{12}	a_{13}	a_{23}	Weights	CR
T2	Delivery, Price, Availability	5	4	1/3	0.674, 0.101, 0.226	8.25%
T3	Aftersales, Quality, Sustain.	1/3	2	3	0.249, 0.594, 0.157	5.16%
T4	Delivery time: S1, S2, S3	1/5	1/3	3	0.105, 0.637, 0.258	3.70%
T5	Part quality: S1, S2, S3	3	7	5	0.649, 0.279, 0.072	6.24%
T6	Availability: S1, S2, S3	1/3	1/3	1	0.143, 0.429, 0.429	0.00%
T7	Aftersales: S1, S2, S3	3	3	2	0.594, 0.249, 0.157	5.16%
T8	Price: S1, S2, S3	1/6	1/8	1	0.067, 0.444, 0.489	0.88%
T9	Sustainability: S1, S2, S3	3	5	3	0.637, 0.258, 0.105	3.70%

Source: answers from Rodrigues et al. (2022), Tables 2 to 9; weights and CR recomputed by the author

3.4 AHP and ANP Models

In the AHP model the two groups are treated as equally important. For ANP the authors describe a network in which the two groups are linked and receive feedback from the suppliers, so the groups are no longer equally important. All calculations were done in the SuperDecisions software.

3.5 Results Reported in the Paper

The authors recommend the ANP answer, Supplier 1, because ANP suits problems where the criteria are not strictly in a tree. They also note two difficulties: the managers

Table 3.2: Final scores reported by the paper

Method	**Supplier 1**	**Supplier 2**	**Supplier 3**	**Winner**
AHP (Table 10)	0.371467	**0.419439**	0.209094	Supplier 2
ANP (Table 11)	**0.546076**	0.318691	0.135233	Supplier 1

Source: Rodrigues et al. (2022), Tables 10 to 12

were unwilling at first to use the methods on confidential data, and keeping the answers consistent was hard because the managers were using the methods for the first time.

Chapter 4

Gaps Identified in the Paper

4.1 How the Paper Was Checked

All eight tables were entered into a program written for this seminar. Weights were found with the power method (multiply the matrix by a vector and repeat until the vector stops changing), which gives the principal eigenvector. The consistency ratio used $RI = 0.52$ for $n = 3$, the value used by SuperDecisions; with it the program reproduces every consistency value printed in the paper. The final score of supplier i is

$$S_i(w) = w \sum_{a \in C1} v_a \, p_{a,i} \; + \; (1 - w) \sum_{a \in C2} v_a \, p_{a,i}, \tag{4.1}$$

where w is the weight of group C1, v_a is the weight of attribute a inside its group and $p_{a,i}$ is the score of supplier i on attribute a. With $w = 0.5$ this gives 0.3715, 0.4194 and 0.2091, exactly the paper's AHP result.

4.2 Gap 1: Two Methods, Two Winners

Equation 4.1 is a straight line in w:

$$S(0) = (0.6334,\ 0.2683,\ 0.0983), \qquad S(1) = (0.1096,\ 0.5706,\ 0.3199).$$

Solving $S_1(w) = 0.546076$ gives $w = 1/6$. At this value the other two scores are 0.3187 and 0.1352, which also match the paper. So the paper's ANP ranking is exactly an AHP ranking in which quality is five times as important as logistics. No link between attributes and no feedback from the suppliers is needed to produce it. The change of winner comes from one judgment about the two groups, which is exactly the point the managers disagree on, and the paper does not show this.

4.3 Gap 2: No Sensitivity Check

Setting $S_1(w) = S_2(w)$ gives the weight at which the winner changes:

$$w^* = \frac{S_1(0) - S_2(0)}{[S_1(0) - S_2(0)] - [S_1(1) - S_2(1)]} = \frac{0.3651}{0.3651 + 0.4610} = 0.442. \tag{4.2}$$

Supplier 1 wins when the logistics group gets less than 44.2% of the weight and Supplier 2 wins above it (Figure 4.1). Supplier 3 is never first. The paper's AHP point ($w = 0.5$) is only 0.058 away from the flip point.

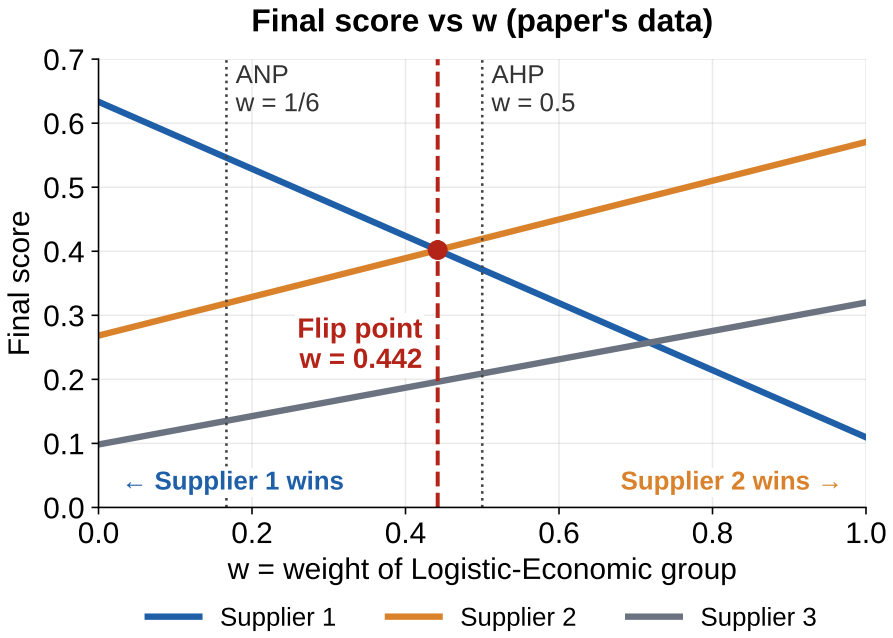


Figure 4.1: Final scores as the weight of the logistics group changes (paper's data)

Source: own calculation with Equation 4.1 from the paper's Tables 2 to 9

Each attribute's weight was also changed alone, with the others scaled so that the total stays 1, to find how far it must move before the winner changes (Table 4.1). Delivery time, part quality, aftersales and sustainability can each flip the decision with a change of only 0.07 to 0.11.

Table 4.1: One-attribute sensitivity at the paper's AHP setting ($w = 0.5$)

Attribute	Weight now	Flips at	Change	New winner
Delivery time	0.337	0.271	−0.066	Supplier 1
Price	0.050	0.834	+0.783	Supplier 3
Availability	0.113	never		
Aftersales service	0.125	0.232	+0.107	Supplier 1
Part quality	0.297	0.378	+0.081	Supplier 1
Sustainability	0.079	0.182	+0.104	Supplier 1

Source: own calculation

4.4 Gap 3: Answers Not Consistent

All eight tables are below the common 10% limit, but four of them (T2, T3, T5 and T7 in Table 3.1) are above the 5% limit that Saaty recommends for three items. The least consistent is T2 (8.25%), the table that sets the weights of delivery time and price.

4.5 Gap 4: Measured Values Given as Opinions

Price, delivery time and defect rate are measured by the company, yet the paper scores them with opinions. On the 1 to 9 scale Supplier 1 was judged 6 and 8 times worse than Suppliers 2 and 3 on price, which gives it a price score of 0.067 against 0.489, a ratio of 7.3. The real price ratio is at most $350/230 = 1.52$. The verbal scale stretched a small difference into a large one. The same happens for delivery time and quality (Section 5.3).

4.6 Gaps 5 and 6: One Fixed Ranking and No Cost View

The paper gives a single ranking. Since every answer is a rounded human opinion, a judgment of 5 could easily have been 4 or 6, and the paper does not test whether the winner would survive such small changes. The ranking also says nothing about money: it does not show the cost of buying, of handling bad rods or of line stops while waiting for a rod.

Chapter 5

Proposed Improvement

5.1 Objectives

The improved model keeps the paper's input values and changes only the way they are used:

1. Re-check the most doubtful answer in each inconsistent table (Gap 3).
2. Use real numbers for the attributes that are measured (Gap 4).
3. Show the flip point to the managers (Gaps 1 and 2).
4. Measure how sure the winner is (Gap 5).
5. Show the cost of each buying plan over two years (Gap 6).

All five are built into one simple web tool (Chapter 6).

5.2 Re-checking Doubtful Answers

For every table above 5%, each single answer was moved by one step on the scale (for example from 3 to 2), and the change that gives the lowest consistency ratio was kept. These are the answers the managers should be asked to look at again. One change per table was enough (Table 5.1, Figure 5.1).

Table 5.1: The one answer to re-check in each inconsistent table

Table	Answer	Change	CR before	CR after
T2	Price vs Availability	$1/3 \to 1/2$	8.25%	2.36%
T3	Aftersales vs Sustainability	$2 \to 1$	5.16%	0.00%
T5	Part quality: S1 vs S2	$3 \to 2$	6.24%	1.36%
T7	Aftersales: S2 vs S3	$2 \to 1$	5.16%	0.00%

Source: own calculation

5.3 Scoring from Real Numbers

For an attribute where a lower value is better and a measured value x_i exists, the score of supplier i is

$$p_i = \frac{1/x_i}{\sum_k 1/x_k}. \tag{5.1}$$

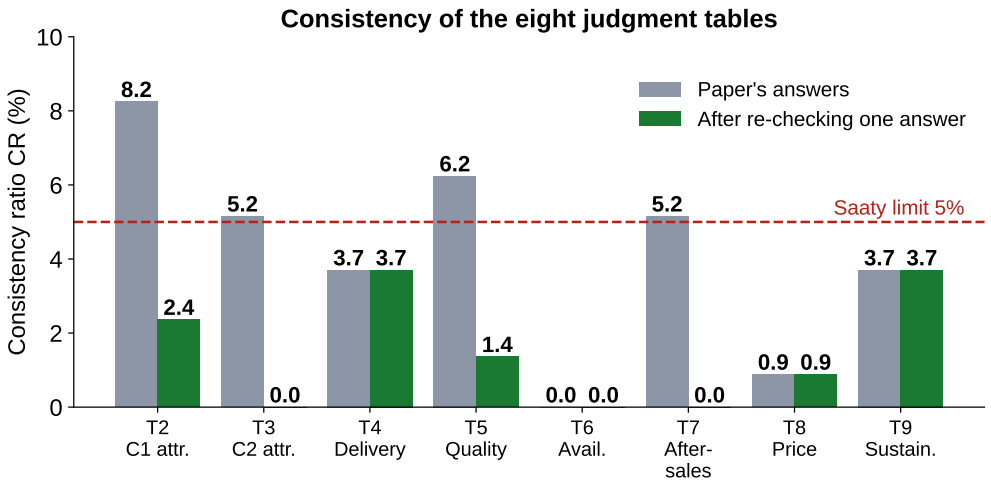


Figure 5.1: Consistency ratio of the eight tables before and after the re-check

Source: own calculation

This keeps the real ratio: a supplier that is 1.52 times dearer gets a price score 1.52 times lower. Availability, aftersales and sustainability keep the managers' answers. The paper gives only ranges, so the value of each supplier was placed inside the paper's range following the paper's own ranking (Table 5.2). The tool lets the user replace them with the company's real figures.

Table 5.2: Measured values used in the improved model

Measure	S1	S2	S3	Paper's range
Unit price (€)	350	230	235	230 to 350
Delivery time (days)	7	3	5	3 to 7
Bad parts 2018 (%)	6.67	10.00	15.00	6.67 to 15

Source: ranges from Rodrigues et al. (2022), Section 3; per-supplier values estimated by the author

5.4 Measuring How Sure the Winner Is

In each of 5,000 trials every answer is moved at random by -1, 0 or $+1$ step on the scale, the group weight w is picked at random between 0.1 and 0.9, and every measured value is varied by up to $\pm 5\%$. The winner of each trial is recorded. A supplier is treated as a safe choice if it wins at least 80% of the trials. A fixed random seed is used so that the result can be repeated exactly.

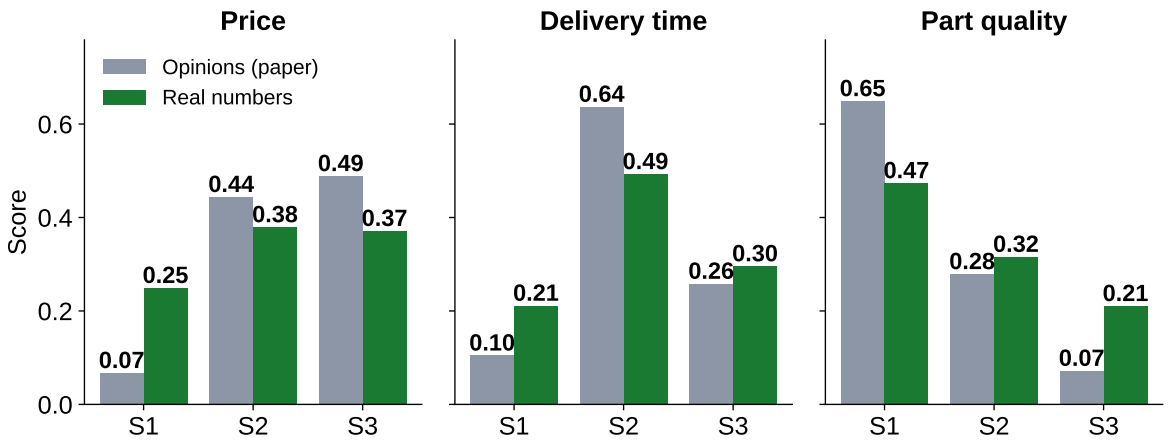


Figure 5.2: Scores on price, delivery time and part quality from opinions and from real numbers

Source: opinions recomputed from paper Tables 4, 5 and 8; real numbers from Table 5.2

5.5 Two-year Cost of Each Buying Plan

The expected cost of four plans (only S1, only S2, only S3, and a 50/50 split between S1 and S2) is worked out month by month over 24 months:

$$\text{Buying cost} = \sum_{t} \frac{R}{12} P_i (1 + r)^{t/12}, \tag{5.2}$$

$$\text{Bad rods} = \sum_{t} \frac{R}{12} \left(d_i + g \, \frac{t}{12} \right), \tag{5.3}$$

$$\text{Stop hours} = (\text{bad rods} + \text{urgent needs}) \times q \times L_i \times h, \tag{5.4}$$

$$\text{Total cost} = \text{buying cost} + \text{bad rods} \times c_b + \text{stop hours} \times c. \tag{5.5}$$

Each bad rod or urgent breakdown is a wait for a new rod. With probability q the spare shelf is empty, and the line then stops for the supplier's delivery time L_i. Table 5.3 lists the inputs and where they come from.

Table 5.3: Inputs of the two-year cost model

Input	Value	Source
Rods needed per year, R	90	Paper: yearly spend / average price
Rise in bad parts, g	1.67 points/year	Paper: rise from 2017 to 2018
Copper price rise, r	5%/year	Estimate (copper market)
Handling cost per bad rod, c_b	€150	Estimate
Urgent needs per year	4	Estimate
Chance shelf is empty, q	5%	Estimate
Hours lost per waiting day, h	8	Estimate (one shift)
Cost of one stop hour, c	€300	Estimate

Source: as stated in the third column

Chapter 6

Web Tool and Results

6.1 The Web Tool

The improved model was built as a single web page called Supplier Decision Lab, written in plain HTML, CSS and JavaScript, with charts drawn by Chart.js. It runs without internet; it starts with one command in the project folder (`node server.js`) and opens at `http://localhost:8080`. The page opens with a summary of the paper's model next to the improved model (Figure 6.1), followed by eight steps. Each step shows the input values and the working, so every number can be checked by hand (Figure 6.2). Any input can be changed, and every result updates at once.

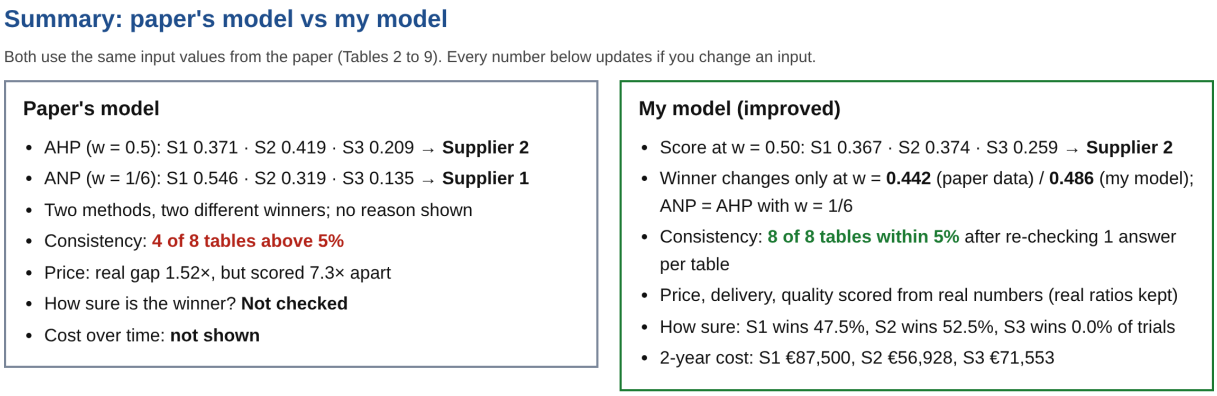


Figure 6.1: Web tool: summary of the paper's model and the improved model

Source: screenshot of the developed tool (own work)

6.2 Effect of Each Improvement on the Scores

Table 6.1 adds the improvements one at a time at equal group weights ($w = 0.5$). Re-checking the answers alone widens Supplier 2's lead. Real numbers alone narrow it, because Supplier 1's price and delivery are no longer exaggerated. With both, Supplier 2 stays first by a small margin of 0.007, and the flip point moves to 0.486 (Figure 6.3).

6.3 How Sure the Winner Is

Table 6.2 and Figure 6.4 give the results of 5,000 trials. Supplier 3 never wins and finishes last in about 77% of trials, so it can be dropped. Neither Supplier 1 nor Supplier 2 comes

Step 3. Final score of each supplier (with working)

Final score = Σ (global weight of attribute × score of supplier on that attribute). Global weight = group weight × weight inside the group.

Weight of group C1 Logistic-Economic, w = 0.500 (C2 Quality-Sustainability gets 1 − w) Paper AHP: w = 0.5 Paper ANP: w = 1/6

Attribute	Group weight	× weight in group	= global weight	Supplier 1 score	Supplier 2 score	Supplier 3 score	S1: weight × score	S2: weight × score	S3: weight × score
Delivery time (C1)	0.500	0.674	**0.3369**	0.105	0.637	0.258	0.0353	0.2146	0.0870
Price (C1)	0.500	0.101	**0.0503**	0.067	0.444	0.489	0.0034	0.0223	0.0246
Availability (C1)	0.500	0.226	**0.1128**	0.143	0.429	0.429	0.0161	0.0483	0.0483
Aftersales (C2)	0.500	0.249	**0.1247**	0.594	0.249	0.157	0.0740	0.0311	0.0196
Quality (C2)	0.500	0.594	**0.2968**	0.649	0.279	0.072	0.1927	0.0828	0.0213
Sustainability (C2)	0.500	0.157	**0.0785**	0.637	0.258	0.105	0.0500	0.0203	0.0082
Final score (sum of each column)							**0.3715**	**0.4194**	**0.2091**
Paper, Table 10 (AHP)							*0.3715*	*0.4194*	*0.2091*

Winner: Supplier 2. Using w = 0.5 gives the paper's AHP result. Using w = 1/6 (quality 5 times as important as logistics) gives exactly the paper's ANP result, so the change of winner comes from this one weight.

Figure 6.2: Web tool, Step 3: final score with every product of weight and score shown

Source: screenshot of the developed tool (own work)

Table 6.1: Final scores at $w = 0.5$ as each improvement is added

Model	S1	S2	S3	Winner	Flip point
Paper's AHP	0.3715	0.4194	0.2091	Supplier 2	0.442
Re-checked answers only	0.3552	0.4322	0.2126	Supplier 2	0.401
Real numbers only	0.3644	0.3786	0.2570	Supplier 2	0.472
My model (both)	**0.3668**	**0.3740**	**0.2592**	**Supplier 2**	**0.486**

Source: web tool, Steps 3, 5 and 6

close to the 80% mark in either model, so the ranking alone cannot decide between them. The improved model makes the two even closer than the paper's model suggests.

Table 6.2: Share of 5,000 trials in which each supplier finishes 1st, 2nd and 3rd

	Paper's model			My model		
Supplier	1st	2nd	3rd	1st	2nd	3rd
Supplier 1	41.9%	34.2%	23.9%	47.5%	30.2%	22.2%
Supplier 2	58.1%	41.9%	0.0%	52.5%	47.5%	0.0%
Supplier 3	0.0%	23.9%	76.1%	0.0%	22.2%	77.8%

Source: web tool, Step 7 (5,000 trials, seed 7)

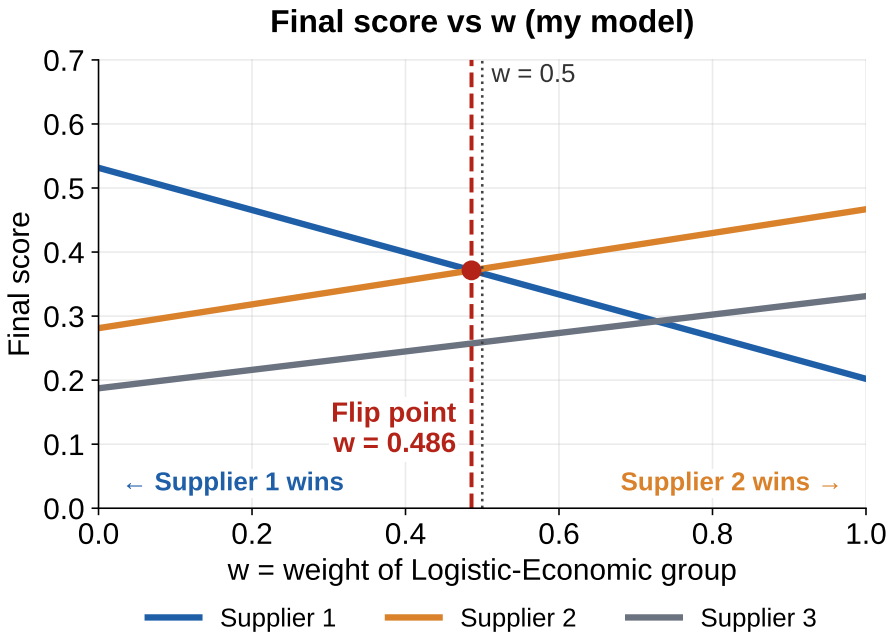


Figure 6.3: Final scores against the weight of the logistics group in the improved model

Source: web tool, Step 6

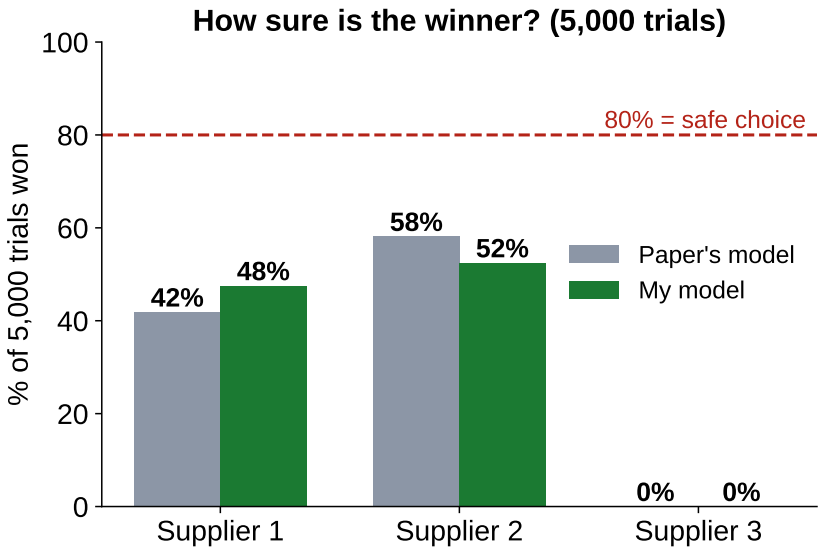


Figure 6.4: Share of 5,000 trials won by each supplier

Source: web tool, Step 7

6.4 Two-year Cost

Table 6.3 and Figure 6.5 give the expected cost of each plan. Buying only from Supplier 2 is the cheapest plan. Supplier 1 has the fewest bad rods, but its higher price and longer delivery from Germany add both buying cost and line-stop hours. The 50/50 split costs about the same as buying only from Supplier 3, but with far fewer bad rods.

Table 6.3: Expected two-year cost of each buying plan

Plan	Buying (€)	Bad rods (€)	Line stops (€)	Total (€)	Bad rods (no.)	Stop (hours)
Only S1	66,042	2,233	19,225	87,500	14.9	64.1
Only S2	43,399	3,132	10,397	**56,928**	20.9	34.7
Only S3	44,342	4,482	22,728	71,553	29.9	75.8
Split 50/50 S1, S2	54,720	2,683	14,811	72,214	17.9	49.4

Source: web tool, Step 8, inputs of Table 5.3

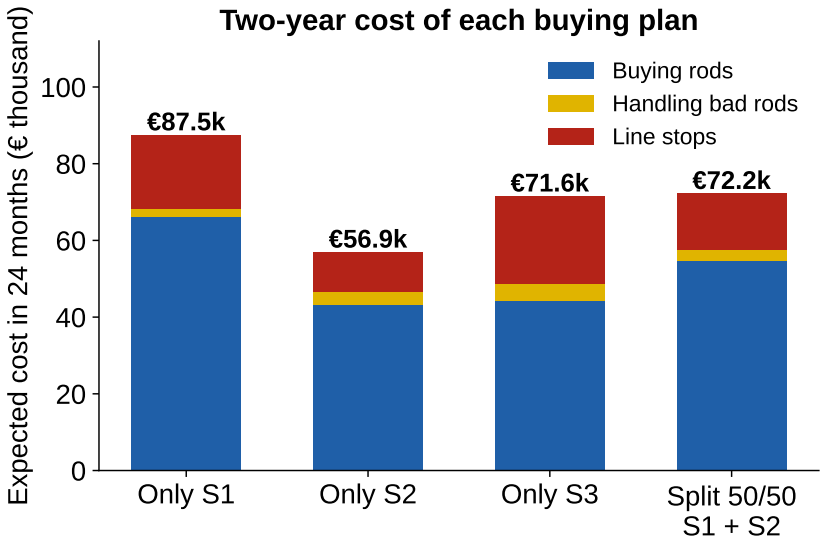


Figure 6.5: Where the money goes in two years for each plan

Source: web tool, Step 8

The tool also turns the cost model into three clear figures for the managers:

- Supplier 1 becomes as cheap as Supplier 2 over two years only if one bad rod costs the company more than about €5,250 (through scrap, a customer complaint or a recall) instead of €150.
- To match Supplier 2 on total cost, Supplier 1 would have to cut its price from €350 to about €188 per rod, a 46% discount.

- Supplier 1 causes about 29 more hours of line stop than Supplier 2, because each wait for a rod lasts 7 days instead of 3.

6.5 Paper's Model Compared with My Model

Table 6.4: Paper's model compared with the improved model

	Paper's model	**My model**
Winner	Depends on the method; no reason given	Supplier 2 just ahead; flip point shown
Consistency	4 of 8 tables above 5%	8 of 8 tables within 5%
Price scoring	1.52 times gap scored as 7.3 times	True ratio kept
How sure	Not checked	Tested with 5,000 trials
Cost view	Not shown	Two-year cost and break-even values
Checking	Hidden inside software	Every step shown in the tool

Source: this chapter

6.6 Discussion

The paper recommends Supplier 1 because ANP chose it. This work shows that the ANP choice comes from one weighting decision, and that the two leading suppliers are almost tied once the answers are made consistent and measured values are used. When a decision is this close, cost should settle it. Supplier 2 is clearly cheaper over two years, and Supplier 1 is worth its price only if a defective rod is very costly. A split order keeps a second source and lowers the number of bad rods at a moderate extra cost. The cost figures rest on estimates that the company can replace in the tool with its own data.

Chapter 7

Conclusion and Future Work

7.1 Conclusion

This seminar reviewed a published AHP and ANP model for choosing a copper rod supplier and rebuilt it from its own tables. Both published results were reproduced exactly, and the ANP result turned out to be an AHP result with the two groups weighted 1 : 5. The paper's single answer hides how fragile the choice is: it flips at a weight of 0.442, half of its tables fail the 5% consistency check, and measured values were distorted by the verbal scale.

The improved model uses the same input values but re-checks one answer per table, scores measured attributes from real numbers, shows the flip point, tests the result with 5,000 trials and adds a two-year cost view. The recommendation to the company is to drop Supplier 3 and buy from Supplier 2, or to split the order 50/50 with Supplier 1 if the cost of a bad rod is high. A decision model should report not only who wins, but also how sure that answer is and what it will cost.

7.2 Recommendation to the Company

Based on this work, the following steps are suggested to the company before it signs a supply contract:

1. **Drop Supplier 3.** It never wins in any trial and has the most bad parts, so it does not need further study.
2. **Ask the managers to re-check four answers.** The answers listed in Table 5.1 are the ones that make the tables inconsistent. A short meeting is enough to confirm or change them.
3. **Agree on the group weight first.** The managers should decide together how much logistics matters compared with quality. If the agreed weight is above about 0.49, Supplier 2 is the clear choice; below it, Supplier 1 should be considered.
4. **Find the real cost of one bad rod.** This single number (scrap, rework, customer complaint) decides whether the better quality of Supplier 1 is worth its price. The break-even value is about €5,250.
5. **Start with Supplier 2 and keep Supplier 1 as a second source.** If the cost of a bad rod turns out to be high, move to a 50/50 split order.

6. **Review the decision after one year** with the real prices, delivery times and defect rates of that year, entered directly into the web tool.

7.3 Limitations of This Work

- The paper gives only ranges for price, delivery time and defect rate, so the value for each supplier is an estimate inside those ranges.
- The re-checked answers are suggestions; the real answers should come from the managers.
- The individual answers of each manager are not published, so their separate influence could not be studied.
- The cost model uses planning estimates for line-stop cost, urgent needs and stock levels.
- The ANP supermatrix is not published, so the ANP model could be matched but not rebuilt step by step.

7.4 Future Work

- Use fuzzy AHP so that each answer can be a range instead of a single number.
- Give managers different weights based on experience and role, using objective methods such as those reviewed by Koksalmis and Kabak (2019).
- Replace the estimates with the company's real yearly data on prices, defects and line stops.
- Find the best order split between suppliers by optimisation, following Alegoz and Yapicioglu (2019).
- Connect the tool to the company's ERP data so that scores and costs update automatically.

References

1. Rodrigues, M., Šírová, E., and Mugurusi, G. (2022). A supplier selection decision model using multi-criteria decision analysis in a small manufacturing company. *IFAC PapersOnLine*, 55(10), 2773-2778. https://doi.org/10.1016/j.ifacol.2022.10.149

2. Saaty, T. L. (1980). *The Analytic Hierarchy Process.* McGraw-Hill, New York.

3. Saaty, T. L. (2005). *Theory and Applications of the Analytic Network Process: Decision Making with Benefits, Opportunities, Costs and Risks.* RWS Publications, Pittsburgh.

4. Alegoz, M., and Yapicioglu, H. (2019). Supplier selection and order allocation decisions under quantity discount and fast service options. *Sustainable Production and Consumption*, 18, 179-189.

5. Kabir, G., and Hasin, M. A. A. (2011). Comparative analysis of AHP and fuzzy AHP models for multicriteria inventory classification. *International Journal of Fuzzy Logic Systems*, 1(1), 1-16.

6. Gajpal, P. P., Ganesh, L. S., and Rajendran, C. (1994). Criticality analysis of spare parts using the analytic hierarchy process. *International Journal of Production Economics*, 35, 293-297.

7. Molenaers, A., Baets, H., Pintelon, L., and Waeyenbergh, G. (2012). Criticality classification of spare parts: A case study. *International Journal of Production Economics*, 140(2), 570-578.

8. Rezaei, J., Fahim, P. B., and Tavasszy, L. (2014). Supplier selection in the airline retail industry using a funnel methodology: Conjunctive screening method and fuzzy AHP. *Expert Systems with Applications*, 41(18), 8165-8179.

9. Koksalmis, E., and Kabak, Ö. (2019). Deriving decision makers' weights in group decision making: An overview of objective methods. *Information Fusion*, 49, 146-160.